

Worksheet — Arithmetic Progressions (Class 10 NIOS Mathematics)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 11 questions. Show all working for the subjective questions. Total time: 50 minutes.

Q1. Which of the following is an AP? [1 mark]

- (A) 1, 4, 9, 16, 25
- (B) 2, 5, 8, 11, 14
- (C) 1, 2, 4, 8, 16
- (D) 1, 1, 2, 3, 5

Q2. If $d = 0$, every term of the AP is the same. [1 mark]

- (A) True
- (B) False

Q3. For an AP with first term a and common difference d , the n th term $a_n = \underline{\hspace{2cm}}$. [1 mark]

Q4. The 10th term of the AP 2, 5, 8, 11, ... is: [1 mark]

- (A) 27
- (B) 29
- (C) 30
- (D) 32

Q5. Which term of the AP 3, 7, 11, 15, ... equals 99? [2 marks]

- (A) 24th
- (B) 25th
- (C) 26th
- (D) 27th

Q6. The 4th term of an AP is 14 and the 10th term is 38. Find the AP. [2 marks]

Q7. Find the sum of the first 20 terms of the AP 5, 8, 11, 14, ... [3 marks]

Q8. Find $2 + 4 + 6 + \dots + 100$. [3 marks]

Q9. Insert 4 arithmetic means between 3 and 23. [3 marks]

Q10. Sita borrows ₹3000 and repays in 12 monthly instalments. The first instalment is ₹500 and each later instalment is ₹25 less than the previous one. Find the total amount she pays back, and the amount of the 12th instalment. [6 marks]

Q11. How many terms of the AP 24, 21, 18, ... must be summed to give 78? Why are there two answers? [7 marks]

Answer Key

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Q1. Which of the following is an AP?

Answer: 2, 5, 8, 11, 14

Differences are 3, 3, 3, 3 — all equal.

MCQ · EASY · 10-NIOS-MATH-AP-01

Q2. If $d = 0$, every term of the AP is the same.

Answer: True

$d = 0$ makes the AP constant.

TRUE_FALSE · EASY · 10-NIOS-MATH-AP-01

Q3. For an AP with first term a and common difference d , the n th term $a_n = \underline{\hspace{2cm}}$.

Answer: $a + (n - 1) \cdot d$

Standard n th-term formula.

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Q4. The 10th term of the AP 2, 5, 8, 11, ... is:

Answer: 29

$a = 2, d = 3; a_{10} = 2 + 9 \cdot 3 = 29$.

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Q5. Which term of the AP 3, 7, 11, 15, ... equals 99?

Answer: 25th

$3 + (n-1) \cdot 4 = 99 \rightarrow n - 1 = 24 \rightarrow n = 25$.

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Q6. The 4th term of an AP is 14 and the 10th term is 38. Find the AP.

Answer: $d = (38 - 14)/(10 - 4) = 4$. $a = 14 - 3 \cdot 4 = 2$. AP: 2, 6, 10, 14, 18, ...

Use $d = (a_n - a_m)/(n - m)$, then $a = a_n - (n - 1)d$.

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Q7. Find the sum of the first 20 terms of the AP 5, 8, 11, 14, ...

Answer: $a = 5, d = 3, n = 20$. $S_{20} = 20/2 \cdot [2 \cdot 5 + 19 \cdot 3] = 10 \cdot (10 + 57) = 670$.

Apply $S_n = n/2 \cdot [2a + (n-1)d]$.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-AP-05

Q8. Find $2 + 4 + 6 + \dots + 100$.

Answer: AP with $a = 2$, $d = 2$. $100 = 2 + (n-1) \cdot 2 \rightarrow n = 50$. $S_n = 50/2 \cdot (2 + 100) = 25 \cdot 102 = 2550$.

Use $S_n = n/2 \cdot (a + l)$.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-AP-06

Q9. Insert 4 arithmetic means between 3 and 23.

Answer: $d = (23 - 3)/(4 + 1) = 4$. AMs: 7, 11, 15, 19. Resulting AP: 3, 7, 11, 15, 19, 23.

n means in $(n+1)$ gaps; $A_k = a + k \cdot d$.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-AP-07

Q10. Sita borrows ₹3000 and repays in 12 monthly instalments. The first instalment is ₹500 and each later instalment is ₹25 less than the previous one. Find the total amount she pays back, and the amount of the 12th instalment.

Answer: AP: $a = 500$, $d = -25$, $n = 12$. 12th instalment = $a_n = 500 + 11 \cdot (-25) = ₹225$. Total paid = $S_n = 12/2 \cdot [2 \cdot 500 + 11 \cdot (-25)] = 6 \cdot (1000 - 275) = 6 \cdot 725 = ₹4350$.

Identify (a, d, n) ; apply a_n for the last term and S_n for the total.

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Q11. How many terms of the AP 24, 21, 18, ... must be summed to give 78? Why are there two answers?

Answer: $a = 24$, $d = -3$. $S_n = n/2 \cdot [2 \cdot 24 + (n-1) \cdot (-3)] = 78 \rightarrow n \cdot (51 - 3n) = 156 \rightarrow n^2 - 17n + 52 = 0 \rightarrow (n - 4)(n - 13) = 0 \rightarrow n = 4$ or $n = 13$. The AP eventually decreases through zero into negative terms; the running total reaches 78 at $n = 4$, dips, and comes back to 78 at $n = 13$.

Both roots are valid; the AP's later negative terms can pull the cumulative sum back to a previous value.

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